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B.Tech. Degree II Semester Regular Examination July 2024

EE/EC/CS/IT 23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++
(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the basic concepts of object oriented programming.
 CO2: Describe the object oriented programming with concepts of streams, classes, functions, data and objects.
 CO3: Implement object oriented programming construct like encapsulation, inheritance and polymorphism.
 CO4: Understand dynamic memory management technique using pointers, constructors, destructors etc.
 CO5: Identify classes including data, methods and the relationship among the classes from a given statement and solve problem using object oriented constructs in C++.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

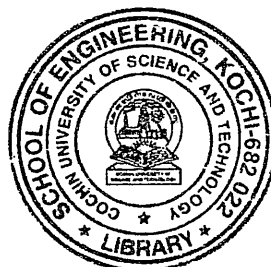
	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a) Differentiate between Constructor and Destructor.	2	2	L2	2	1,2
(b) Explain the use of 'this' pointer.	2	2	L1	1	1
(c) What is the use of new and delete operators?	2	2	L1	2	3,12
(d) List the operator that can't be overloaded.	2	2	L2	3	3,12
(e) Differentiate between Sequential and Random access files.	2	2	L2	4	1,2,3

PART B

(4 × 10 = 40)

II. (a) Explain basis concepts of OOPs.	5	5	L1	1	1
(b) Write a program to check whether a character is a vowel or consonant.	5	5	L1	1	1
OR					
III. (a) Write program to find factorial of a number.	5	5	L1	1	1
(b) Difference between procedural oriented programming and object oriented programming.	5	5	L1	1	1

(P.T.O.)



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		Marks	BL	CO	PO
IV.	(a) Write a program to print the names of student by creating a student class. If no name is passed while creating an object of the student class, then the name should be "unknown", otherwise the name should be equal to the string value passed while creating the object.	5	L2	4	1,2
	(b) Explain friend function in detail.	5	L2	4	3,12
OR					
V.	(a) Explain the different types of constructors with example.	5	L2	3	1-3
	(b) Predict the output and explain it.	5	L2	2	1-3

```
#include <iostream.h>
using namespace std;
class A
{
    int x;
public:
    A(int d)
    { x=d;
    }
    ~A()
    { cout<<x;
    }
};
int main() {
    A a(50),b(60);
    for(int i=1;i<=10;i++)
    { A y(i);
    }
    A c(70);
    for(int i=11;i<=20;i++)
    { A z(i);
    }
    return 0;
}
```

(Continued)

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		Marks	BL	CO	PO
VI.	(a) Unary operators overloaded by means of a member function take no explicit arguments and return no explicit values. Justify your answer with program.	5	L3	3	1
	(b) Explain types of Inheritance with an example.	5	L3	3	1
OR					
VII.	(a) Identify the inheritance and write a program exhibiting the inheritance.	5	L3	3	1
	(b) We have two classes Time and Minute. If 'm' is an object of Minute Class and 't' is an object of Time Class and we want convert Time Class to Minute Class. Produce the conversion routine with a suitable program.	5	L3	3	1
VIII.	(a) What is the advantage of virtual functions? Explain with a suitable program.	5	L2	4	1-3
	(b) Write a C++ program to create a new text file and write some text into it.	5	L3	4	1-3
OR					
IX.	(a) Explain the types of templates used in C++.	5	L3	4	1-3
	(b) Explain under what circumstance the following statement would be used. Write a sample code.	5	L2	4	1-3
	(i) throw				
	(ii) catch(...).				

Blooms's Taxonomy Levels

L1 – 28.57%, L2 – 42.85%, L3 – 28.57%.
